

Convex Core — A Derivation from First Principles

How the allocation equation $w_{\text{eq}} + w_{\text{cvx}} + w_{\text{dur}} = 1$ and the volatility brake follow from one objective: maximizing long-horizon compound growth.

What this is — and isn't. Convex Core was originally *authored from empirical evidence* (the 2026-06-11 design spec, built from the day's backtests). This note shows that the *same* equation is what a first-principles argument recommends — so the theory and the empirics agree, which is the strongest position to be in. It is a principled justification, **not a uniqueness proof**: the derivation terminates in a small number of named design choices, and we mark each as **rigorous** or **design choice** so the reader can see exactly where judgment enters.

Step 0 — The objective: maximize *geometric growth*

Terminal wealth is multiplicative, so the correct objective is not the arithmetic mean return but the expected log-growth rate:

$$g = \mathbb{E}[\ln(1 + R)]$$

long-run growth

To second order (a Taylor / Itô expansion in R):

$$g \approx \mu - \frac{1}{2} \sigma^2$$

$\mu = \mathbb{E}[R]$, $\sigma^2 = \text{Var}(R)$

This one line drives the entire model. Variance is a *direct subtraction* from compound growth. Reducing σ^2 raises g *even if it lowers* the arithmetic mean μ — so "give up $\leq 1\%$ /yr of arithmetic return to cut risk" is not risk-aversion, it is **growth-optimal**. **rigorous**

Drawdowns sharpen the point. Recovering from a loss of size d requires a gain of

$$d / (1 - d) > d$$

–50% needs +100%

so recovery is **convex** in the loss, and negative skew (left-tail losses) makes the realized drag *worse* than the symmetric $\frac{1}{2}\sigma^2$ term suggests. The quantity to minimize is therefore not total variance but specifically the **left-tail / drawdown variance**.

Step 1 — Why a *structural* allocation, not Markowitz

Classical mean–variance optimization solves for weights from (μ, Σ) . It requires estimating expected returns μ — and the project's first empirical finding is that **μ is not forecastable** at useful precision on this universe (market-timing search returned negative; cross-sectional information coefficient ≈ 0.026). Under estimation error, mean–variance optimization is "error-maximizing" (Michaud): tiny μ -errors produce wildly unstable weights.

The principled response is to **stop forecasting returns** and allocate by *role*, under the one constraint that needs no estimate — full investment, no leverage:

$$\sum_i w_i = 1$$

the budget constraint

That is the left-hand side of the equation. The only open question is *what the roles are*.

rigorous, given the premise that μ is unforecastable

Step 2 — The three roles fall out of the risk decomposition

Exactly three functions are needed:

1. **Capture the one premium you believe in.** The equity risk premium is the only large, persistent, theory-backed expected return $\rightarrow$ an **equity core**, w_{eq} .
2. **Hedge the left tail — but the tail has two regimes.** The stock–bond correlation is *regime-dependent*: Treasuries rally in growth-scare / deflationary crashes (2008, 2020) but fall *with* equities in inflation / rate-shock selloffs (2022). One hedge cannot cover both. The tail-hedge therefore splits into two orthogonal sleeves:
 - **Duration**, w_{dur} — carry/ballast that hedges the *deflationary* crash.

- **Crisis-convexity**, w_{cvx} — a long-volatility, trend-following / anti-beta sleeve that hedges the *inflationary / momentum-crash* regime duration misses.

Three terms, summing to one:

$$w_{eq} + w_{cvx} + w_{dur} = 1$$

$$w_{dur} = 1 - w_{eq} - w_{cvx} \text{ (the plug)}$$

Why "convex." Trend-following has an *option-like, convex payoff* — Fung & Hsieh (2001) showed it replicates *long lookback straddles*. Its return curve versus the equity factor **bends upward** in the tails, injecting the positive co-skew that directly attacks the left-tail variance from Step 0. The sleeve is the portfolio's "long-gamma" leg.

design choice The principle *forces* "one premium sleeve + tail diversification"; choosing exactly **two** hedge sleeves for the two documented crash regimes is motivated by — not uniquely implied by — the argument. The 2022 attribution (convexity **+4.5** while duration **-1.6**) is the empirical confirmation that the two sleeves are genuinely orthogonal.

Step 3 — The volatility brake from the Kelly criterion

For the equity core, the log-optimal (Kelly) fraction in a risky asset with excess return $X \sim (\mu_x, \sigma_x^2)$ is

$$f^* = \mu_x / \sigma_x^2$$

Kelly / log-optimal sizing

Now apply the robust-estimation insight again. You cannot estimate μ_x , but the **Sharpe ratio** $S = \mu_x / \sigma_x$ is far more stable, and — crucially — **volatility itself is forecastable** (volatility clustering; the most robust regularity in daily returns). Substituting S :

$$f^* = S / \sigma_x \propto 1 / \sigma_x$$

hold risk constant, not dollars

i.e. scale exposure inversely with realized volatility. Impose the no-leverage constraint ($f \leq$ the strategic weight) and you get a cap at 1 — exactly the brake:

$$eq_scale = \min(1, \sigma_{target} / \hat{\sigma}_{21d})$$

the volatility brake

The brake is the *one* piece of timing that survives, because it conditions only on $\hat{\sigma}$ (estimable from volatility clustering) and never on μ (which is not). It "responds in days, with no classification machinery." **rigorous, under "Sharpe more stable than the mean"**

Step 4 — Freezing the parameters is implied, not bolted on

Because μ is unforecastable (Step 1), *any* in-sample optimization of the weights (w_{eq} , w_{cvx} , σ_{target} , ...) would fit noise and re-introduce exactly the instability the role-based design escaped. The growth-robust choice is to **freeze the five parameters at structurally reasonable values and let the structure, not the calibration, carry the result** — which makes the 21-year backtest out-of-sample by construction. "Zero fitting" is not a footnote; it is *required* by the same premise that produced the equation. **rigorous, given the premise**

the specific values 0.65 / 0.15 / 0.20 are a design choice

Rigorous vs. design choice — at a glance

Result	Status
$g \approx \mu - \frac{1}{2}\sigma^2 \Rightarrow$ minimize (left-tail) variance for growth	rigorous — a math identity
μ unforecastable $\Rightarrow$ abandon mean–variance; allocate by role under $\Sigma w=1$	rigorous — given the empirical premise
Kelly $\Rightarrow$ vol-targeting $\min(1, \sigma_{target}/\hat{\sigma})$	rigorous — under "Sharpe > mean in stability"
Three sleeves specifically (two crash regimes)	design choice — motivated, not forced
The exact frozen weights 0.65 / 0.15 / 0.20	design choice — "freeze," not "freeze here"

The defensible one-liner. The Convex Core equation is the form that *maximizing expected log-growth* implies once one accepts that **expected returns are not forecastable but volatility is**: a permanent premium-capturing core, a convex crisis hedge and a duration ballast for the two crash regimes, all sized by a Kelly-style volatility brake, with parameters frozen because optimizing them would re-create the very estimation error the structure exists to avoid.

Provenance & references

The mechanisms recombine well-published premia, and the general combination — equity + trend crisis-alpha + bond ballast, de-risked by diversification and volatility scaling — is well established (Schwalbach et al. 2025; the return-stacking literature; Standpoint; AQR defensive equity). The paper claims **no novel strategy**. To the authors' knowledge the specific *unlevered, fixed-weight* three-sleeve formulation with this realized-volatility brake and frozen parameters (and no option premium) has not been published as an integrated rule set; the distinct contribution is the integration and the research process, not the formula.

- Kelly, J. L. (1956). *A New Interpretation of Information Rate*. — log-optimal growth.
- Markowitz (1952); Michaud (1989), *The Markowitz Optimization Enigma* — estimation error in MVO.
- Moreira & Muir (2017), *Volatility-Managed Portfolios*; Harvey et al. (2018) — the volatility brake.
- Hurst, Ooi & Pedersen (2017), *A Century of Evidence on Trend-Following* — crisis alpha.
- Fung & Hsieh (2001), *The Risk in Hedge Fund Strategies* — trend-following as long lookback straddles (convexity).
- Frazzini & Pedersen (2014), *Betting Against Beta* — the anti-beta / defensive sleeve.

DNSR Investments, LLC · companion to the Convex Core white paper (§4 Model Philosophy). Hypothetical/backtested context; not investment advice. The portfolio math is fully deterministic — no large-language model participates in the allocation engine.